INKLOOM AI

Intelligent Multi-Brand Brand-Governance Platform

Round 3 Final Master Architecture, Implementation & Production Acceptance Report

Status:	Round 3 Final Production Freeze
Live Web App:	https://inkloom-ai-wheat.vercel.app/
Backend API:	https://inkloom-backend-upau.onrender.com
Database:	AWS PostgreSQL (Supabase Session Pooler)
Repository:	shatakshi-rawat-130613/inkloom-ai (main)
Commit Hash:	0027e50 (ancestor of ba4a0db & 9a45f90)
Automated Tests:	146 passed backend tests (pytest) | 27 live acceptance journeys passed

Executive Summary: Inkloom AI transforms marketing compliance and brand governance from a manual, error-prone bottleneck into an automated, verifiable, and explainable collaborative workflow. In Round 3, the application successfully graduated from a single-brand demonstration into an enterprise multi-brand platform. All three committed improvements have been implemented, deployed, and rigorously verified against live production environments. The system enforces human authority over final approval decisions while providing deep semantic and visual inspection of claims, tone, and brand identity.

1. The Three Committed Improvements

Improvement 1: Multi-Brand Management & Tenant Isolation

Inkloom AI supports dynamic multi-organisation and multi-brand hierarchies with cryptographic tenant isolation. Organisations can register independently, automatically appointing the founding user as Brand Manager (admin). Within an organisation, managers can create multiple brand workspaces (e.g. LumaCare skincare, Botanica Derma, or Finora fintech), each maintaining independent compliance rules, verified factual claims, approved terminology, and visual brand assets.

- **Strict Role-Based Access Control (RBAC):**

- *Brand Manager (admin):* Full management of brand workspaces, knowledge entries, visual assets, and team invitations.

- *Content Creator (editor):* Submits content drafts, uploads marketing creative, generates revisions, and runs evaluations. Prevented from mutating brand rules or inviting users.

- *Compliance Auditor (reviewer):* Reviews queued submissions, grants authoritative approval, rejects non-compliant copy, or requests revisions with mandatory feedback.

- **Cryptographic Tenant Isolation:** Database queries enforce organization boundaries. Unauthorised access attempts return 404 Not Found to prevent leaking tenant resource existence.
- **Alembic Schema Migrations:** Database evolution is governed by versioned Alembic migrations (20261111_01_round3_multibrand_rbac.py), preserving all historical submissions and reviews.

Improvement 2: Multi-Format Ingestion & Genuine Multimodal Analysis

Content analysis is no longer confined to raw text entry. The ingestion engine accepts Word documents (.docx), native text PDFs, scanned image PDFs, plain text, and marketing creatives (.png, .jpg, .webp) up to 10MB.

- **Two-Tier PDF Extraction with Scanned OCR Fallback:** Native text streams are extracted with pypdf. If zero native text is detected (scanned paper copies), the system extracts embedded raster pages, triggers Gemini Vision OCR, tags the source as 'ocr', and calculates exact word counts and page counts.
- **Genuine Visual Logo Verification:** Unlike naive text lookups, the system performs visual reference inspection against uploaded brand assets. Multimodal vision and difference perceptual hashing (dHash) distinguish 4 clear states:

1. *APPROVED_LOGO_DETECTED:* Authentic brand wordmark or logo verified in geometry and expected primary color.
2. *POSSIBLE_INCORRECT_LOGO:* Altered, distorted, unapproved variant, or competing brand logo detected.
3. *EXPECTED_LOGO_NOT_DETECTED:* Missing brand mark on promotional marketing creative.
4. *UNABLE_TO_DETERMINE:* Low-resolution or corrupted imagery preventing reliable determination.

- **Color Palette & Standards Compliance:** K-Means clustering extracts dominant image colors, computing Euclidean and CIELAB distance against approved brand palette swatches.

Improvement 3: Immutable Governance Lifecycle, Revisions & Audit Trail

To satisfy corporate and legal compliance standards, Inkloom AI implements an append-only revision architecture. Submissions transition through an immutable version tree (v1, v2, v3). Revisions never overwrite existing data.

- **Audit-Proof Knowledge Snapshotting:** Every analysis run serializes an immutable snapshot of all active guidelines, facts, and terms at the exact moment of evaluation. Even if rules evolve years later, past compliance verdicts remain explainable and auditable.
- **Human Authority Preservation:** AI analysis functions as decision support. Human auditors hold sole authority over final publication decisions. Approving a non-compliant run is permitted under human sovereignty but requires mandatory explanatory comments.
- **Chronological Audit Logging:** All operations (creation, revision, analysis, review request, review decision) are committed to append-only audit_logs and survive browser refreshes.

2. Technical Architecture & Database Schema

Inkloom AI is structured as a modular monolith. Deterministic Python validation gates wrap raw multimodal model calls, ensuring that language model hallucinations cannot alter compliance status or invent non-existent rule citations.

Component	Technology	Key Architectural Role
Frontend Client	React 19, Vite, Tailwind	Editorial UI, multi-brand selector, document uploader, review workflow controls.
Backend API	FastAPI, Python 3.13	JWT authentication, tenant scoping, validation pipelines, REST endpoints.
Database	PostgreSQL (Supabase AWS)	Alembic versioned migrations, foreign key constraints (ON DELETE RESTRICT).
Multimodal AI	Gemini 3.6/3.8 Flash	Structured JSON extraction, vision OCR, visual logo verification with flash-lite fallback.
Perceptual Engine	Pillow, dHash, K-Means	64-bit difference hashing, Hamming distance, dominant hex palette extraction.
Document Service	pypdf, python-docx	Multi-format structural ingestion, scanned PDF image extraction, Word XML parsing.

3. Production Verification & Testing Evidence

To ensure uncompromising reliability, the entire product was validated across local and live production environments:

Verification Domain	Test Suite / Script	Execution Scope & Result
Backend Automated Tests	backend/tests/ (pytest)	146 tests passed in 16.55s. Full contract parity, models, and rule engine.
Frontend Quality Checks	tsc -b && vite build && eslint	0 TypeScript compilation errors, 0 ESLint warnings, production bundle built.
27-Step Live Acceptance	scripts/live_acceptance_audit.py	100% passed on live Render backend. Registration, brands, reviews, DOCX.
Live Promotional OCR	scripts/live_ocr_and_logo_gap_closure.py	Passed. Extracted visible marketing text verbatim from PNG; tagged source='ocr'.
Live Scanned PDF OCR	scripts/live_ocr_and_logo_gap_closure.py	Passed. Zero native text PDF extracted via page raster fallback; word count: 22.
Live Visual Logo Inspection	scripts/live_ocr_and_logo_gap_closure.py	Passed all 4 states: approved mark (0.98), competing logo, missing, and blurry.
Tenant Isolation Audit	scripts/live_ocr_and_logo_gap_closure.py	Passed. Cross-tenant reads and injections rejected with 404; non-admin invites 403.
Context Sufficiency	scripts/live_ocr_and_logo_gap_closure.py	Passed. 'hellloooooooo' classified as needs_review with insufficient_context.
Violation vs Unsupported	scripts/live_ocr_and_logo_gap_closure.py	Passed. Guarantee flagged as confirmed violation; unapproved claim as unsupported.

4. Security Hardening Summary

- **Zero Hardcoded Fallback Passwords:** Hardcoded demo password defaults have been purged from backend config and frontend bundles. Seeding strictly reads DEMO_SEED_PASSWORD from environment variables.
- **Cryptographic JWT Signature:** Signed using HS256 with strong JWT_SECRET_KEY, enforcing token expiration and organization scoping.
- **RBAC Enforcement:** All brand mutation endpoints (@router.post, patch, delete) require require_admin dependency. Non-admin users attempting privileged calls receive 403 Forbidden.
- **Tenant Leakage Prevention:** Querying or mutating resources belonging to another organization yields 404 Not Found, never confirming resource existence.